

Supplementary Material S4

Complete Stratified Synthesis of the Corpus

Facial Emotion Recognition through Machine Learning in Children with Autism Spectrum Disorder: A systematic scoping review of open-access computational evidence and of the gaps toward therapeutic monitoring

This document complements the table in the manuscript presenting the stratified synthesis restricted to the 11 studies of greatest translational traceability (Section 6.3, Table 17). Here the complete set of studies is reported in two sections: Section A presents the 24 studies of the eligible corpus organised by clinical domain (D1–D4), and Section B presents the 7 studies of contextual evidence outside the eligible set. The additional columns of population, architecture and eligibility group, which do not fit in the printed version, are included here.

The numbering of the # column corresponds to the internal order of the corpus, identical to that employed in Supplementary Material S2, and not to the bibliographic reference numbers of the manuscript. The 3 retracted studies were excluded from the corpus following the editorial status verification described in Section 2.7 of the manuscript and do not appear in this table.

Section A. Eligible corpus (24 studies, organised by clinical domain)

#	Study	Clinical task	Dataset	Population	Setting	Architecture	Metrics	Data prov.
Domain D1: Automated facial expression (8 studies)								
1	Johnson et al. (2025)	Emotion recogn.	AffectNet + clinical	ASD children (N=48)	Lab./Ecol.	EfficientNet-v2	ACC 67.8%	Clinical
8	Radočaj & Mart. (2025)	Emotion recogn.	FERAC	ASD children	Lab.	Swin Transformer	ACC 80%, F1 78.9%	Partial prov.
9	Washington et al. (2022)	Emotion recogn.	Clinical + CAFE (Sub. A)	Children	Lab./Ecol.	ResNet-152	Bal.ACC 79.1%, F1 78%	Clinical
10	Sholikah et al. (2024)	Real-time recogn.	FER-2013	N=3 ASD (test)	Mobile app	VGG-16	ACC 91%, F1 91%	Public general
21	Aljbori et al. (2025)	Emotion recogn.	Kaggle	ASD children	Lab.	3D-CNN + LSTM	ACC 99.82%, F1 98.6%	Web
23	Vedantham et al. (2025)	Emotion recogn.	Kaggle	Children	Lab.	GWO-GEfficientNet	ACC 96.5%, F1 96.4%	Web
27	Dapogny et al. (2019)	Facial expr. quality	JEMImE (N=157)	Children 6–11	Clin. game	Rand. Forests + PCRF	ACC ~82%	Public curated
28	Leo et al. (2018)	Facial prod. ASD	CK+ + clinical (N=17)	ASD children	Clinical	HOG + SVM + CNN	TP 85% (CK+)	Clinical
Domain D2: Affect and therapeutic engagement (5 studies)								
4	Lecciso et al. (2021)	Therapeutic interv.	Private (N=24)	ASD children	Therapy	OpenFace	<i>g</i> =0.52	Clinical

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#	Study	Clinical task	Dataset	Population	Setting	Architecture	Metrics	Data prov.
13	Hassan et al. (2025)	Therapy monit.	DREAM (N=61)	ASD children 3–6	Robot th.	RF + XGBoost + ExtraT.	ACC 95.6%, F1 92.8%	Public curated
15	Rudovic et al. (2018)	Affect / therapy	MIT private (N=35)	ASD children	Robot th.	PPA-net	ICC ~60%	Clinical
16	Cicceri et al. (2026)	Pointing gestures	Clinical private	ASC children	Clinical	4-level framework	mAP 88.6%	Clinical
17	Rudovic et al. (2019)	Active learning	MIT private (N=35)	ASD children	Robot th.	4 LSTMs + Deep RL	Fig.	Clinical
<i>Domain D3: Emotion recognition intervention (3 studies)</i>								
2	He et al. (2019)	Eye-tracking	Private (N=42)	ASD preschoolers (n=21)	Clinical	Eye-tracking	$r=0.619$	Clinical
3	Reed et al. (2021)	Epidemiological	ALSPAC (N=3,562)	Paediatric cohort	Population	Regression	$b=-0.18$	Cohort
18	Almeida et al. (2019)	Serious game	No ML dataset	ASD 6–12	Game	RPG (no ML)	ISO/IEC usab.	N/A
<i>Domain D4: Screening or diagnosis of ASD (8 studies)</i>								
5	Shaik et al. (2025)	ASD detection	Kaggle ASD	Children	Lab.	Ensemble CNN	ACC 84%, F1 81.9%	Web
6	Ahmad et al. (2024)	ASD detection	Kaggle ASD	Children	Lab.	ResNet-50	ACC 92%	Web
12	Sariyanidi et al. (2025)	Autism pred.	CHOP clinical (N=42)	ASD + TD	Clinical	Dict. Learn. + SVM	Fig.	Clinical
14	Alsubai (2025)	ASD detect. (tab.)	Kaggle screening	1,236 months	Screening	Transformer MDNCT	ACC 94.7%	Web
19	Alam et al. (2025)	ASD detection	Kaggle + UIFID	Children	Lab.	ASD-UANet	ACC 96%	Web
20	Hwang et al. (2024)	DD screening	Private (N=89)	34–77 months	Mobile app	LSTM + 2D-FAN	ACC 88%, F1 80%	Clinical
24	Hosney et al. (2024)	ASD detection	Kaggle (co-author)	Children	Lab.	YOLOv8 + DCNN	ACC 97.2%, F1 93%	Web
29	Akın-Bülbül et al. (2024)	ET screening	Private ET	Infants 18–36m	Clinical	KNN + features	ACC 81.45%	Clinical

Section B. Contextual evidence outside the eligible set (7 studies)

The following seven studies do not form part of the eligible corpus and do not participate in any denominator of the manuscript. Each study’s reason for classification as contextual evidence is stated in Section 2.4 of the manuscript: four infringe CE(3) by employing exclusively non-facial modalities and three infringe the age criterion by including exclusively adult participants.

#	Study	Clinical task	Dataset	Population	Setting	Architecture	Metrics	Data prov.	Reason
C1	El Rhatassi et al. (2024)	Emotion recogn.	FER-2013 + Kaggle autism	Autistic adults	Lab.	CNN vs ViT	ACC 65–74%	Public gen.	Adult participants
C2	El Rhatassi et al. (2025)	Real-time recogn.	CK+ + HeyGen AI	Autistic adults	Lab.	CNN×3 + LSTM	ACC val. 53%	Mixed	Adult participants
C3	Levy et al. (2024)	Wearable system	FER-2013	ASD adults (N=10)	Ecological	CNN + HW wearable	Usability	Public gen.	Adult participants
C4	Nagae & Lee (2022)	Therapy + EDA	Private EDA (N=11)	DD children	Robot th.	EDA + wavelet + SVM	Conf. matrix	Clinical	EDA only (CE3)
C5	Papaioannou et al. (2021)	EEG cogn. load	EEG (N=63)	ASD + ADHD adults	Clinical	MSE + PLSC	ANOVA	Clinical	EEG only (CE3)
C6	Xu & Ju (2023)	VBM brain	ABIDE (N=203)	ASD 8–12 (n=98)	Clinical	VBM	GMV-ADOS corr.	Public curated	sMRI only (CE3)
C7	O’Sullivan et al. (2023)	Speaker diariz.	Clinical trial audio (N=72)	ASD 5–45	Home	Pyannote + x-vector	F1 76.5%	Clinical	Speech only (CE3)

Notes.

1. The column # uses the internal corpus order (identical to Supplementary Material S2) for eligible studies and a C-prefix for contextual studies. These numbers do not correspond to the bibliographic reference numbers in the manuscript.
2. The three retracted studies (Güngör et al., 2025; Mahmood et al., 2025; Kumar et al., 2024) were excluded from the corpus following the editorial status verification described in Section 2.7 of the manuscript and do not appear in this table.
3. “Fig.” in the Metrics column indicates that the study reports its performance metrics in graphical format only, without numerically extractable values. This applies to Rudovic et al. (2019) and Sariyanidi et al. (2025).
4. “Bal.ACC” for Washington et al. (2022) refers to balanced accuracy (79.1%), reported on the CAFE subset with human agreement $\geq 60\%$; the value on the full CAFE set is 66.9%.
5. The indication “(CK+)” for Leo et al. (2018) signals that the true positive rate of 85% corresponds to validation on the CK+ public dataset of typically developing adult participants, not to the subsequent application to the 17 children with ASD.
6. “(co-author)” for Hosney et al. (2024) indicates that the Kaggle dataset was created by a co-author of the study, constituting a documented conflict of interest.
7. The “(test)” annotation for Sholikah et al. (2024) indicates that the 3 ASD participants were involved in a subsequent usability test and not in the metric computation.
8. Domain D3 includes Almeida et al. (2019) as a systems design study without ML classification; its metrics correspond to ISO/IEC usability evaluation and are not comparable with supervised classification metrics.

9. The contextual studies in Section B were appraised with the MMAT in Supplementary Material S2 (Section B) but their results do not participate in the aggregate distribution reported in the manuscript's Figure 2.